Pipeline Development for Strategy 2

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21/07/2026

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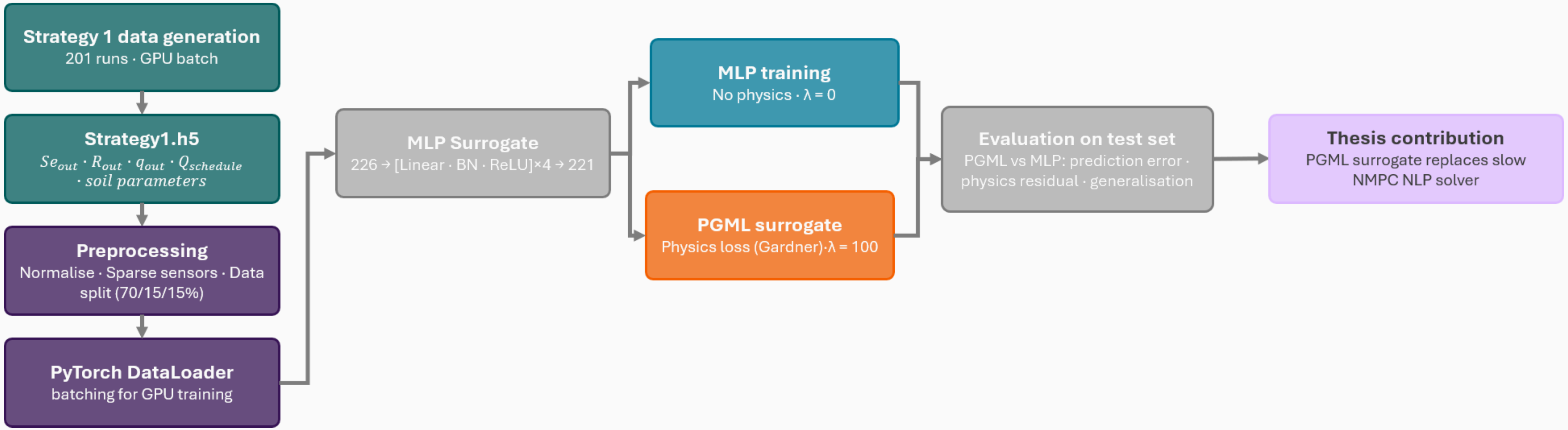
Recap, and what Strategy 2 asks

Strategy 1 built the pipeline; Strategy 2 stresses it

Strategy 1 delivered the training pipeline and a PGML surrogate that replaces the slow NMPC forward solve.



Strategy 2 stresses that surrogate. This deck:

- Data design
- Penalty weight, revisited
- Open loop
- Closed loop
- Speed

Strategy 1 training pipeline (block flow diagram). The slides that follow cover only what changed in Strategy 2.

Strategy 2 design

Structured coverage of the two inputs the controller varies

- **Vary only the control inputs**
  Initial moisture and the irrigation schedule. Everything else stays at Cariaga nominal.
- **Isolate the cause**
  With soil fixed, any change in drift is attributable to coverage, not soil.
- **Space-filling and reproducible**
  512 runs from a scrambled Sobol sequence.

Fixed · Cariaga nominal

Soil: Ks 170 cm/day · α 0.035 1/cm · n 2.267 · θ_s 0.33

Geometry: H 5.5 m · area 308 m^2 · 25 days

Sampled · N = 512 scrambled Sobol

θ_i 0.10 to 0.25 (Se 0.30 to 0.76)

$\bar{Q}$ 0.5 to 48 m^3/day, target irrigation

σ 0 to 1 (0 pins flat at $\bar{Q}$, 1 wanders widely)

Sampled with a scrambled Sobol sequence; soil and geometry held fixed. Source: S2 digest, Sealed Design.

Q-schedule generation

A bounded daily walk around a sampled target

$$Q_0 = Q_{\mathrm{lb}} = 0.5$$

$$lo = \max(\, Q_{\mathrm{lb}},\, (1-\omega)\, Q_{d-1}\,)$$

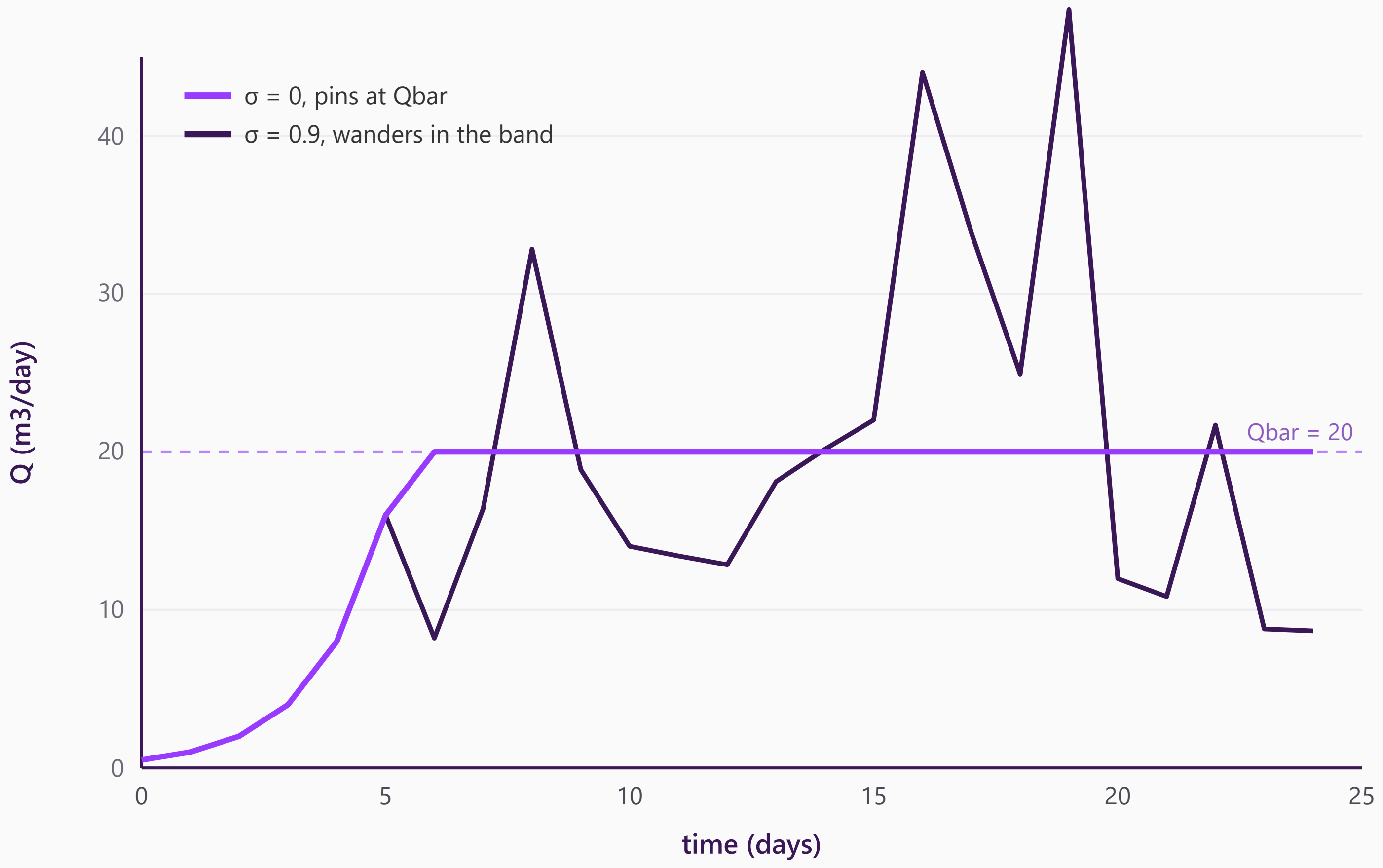
$$hi = \min(\, Q_{\mathrm{ub}},\, (1+\omega)\, Q_{d-1}\,)$$

$$Q_d = \mathrm{clip}(\, \overline{Q} \cdot e^{\sigma u},\, lo,\, hi\,)$$

$$\text{with} \quad u \sim U(-1, 1)$$

- **Bounded daily step**
  The rate stays within (1 ± ω) of the previous day, so moves scale with the rate. With ω = 1 it at most doubles going up, and can drop to the floor in one step.

- **Sampled around a target**
  Each day proposes a random multiple of the target, then clips into the band. σ (0 to 1) sets the spread: 0 pins at the target, 1 wanders widely.



Generator sealed 2026.07.17. The target is the level the schedule is drawn around, not the realised mean; the ramp from the floor and clipping pull the mean a little below it.

Penalty weight, revisited

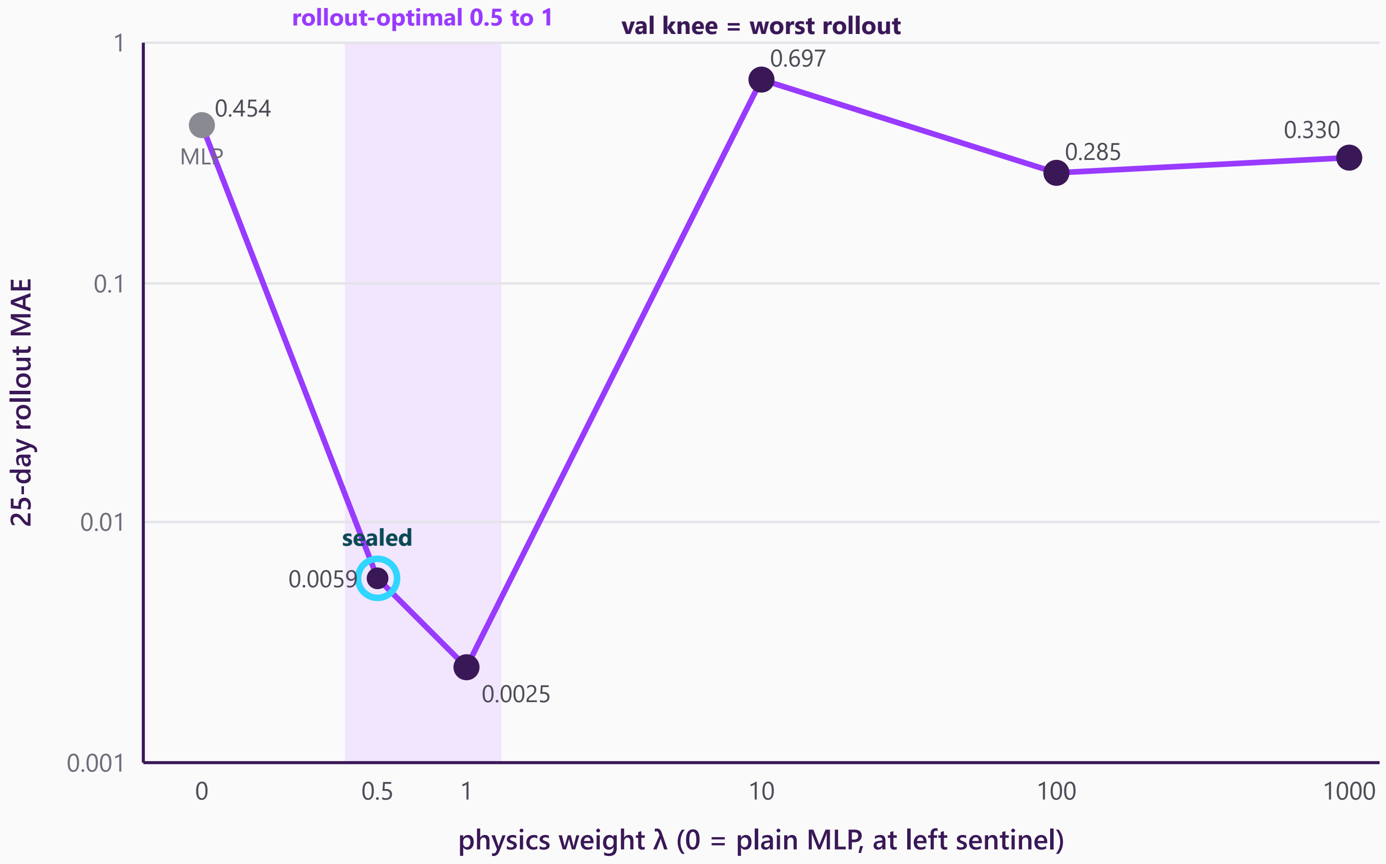
Chosen on 25-day rollout error, not one-step fit

$$L = \text{data error} + \lambda \times \text{physics}$$

λ weights the physics term during training

- **The reversal**
  The penalty weight that looks best on one-step validation is among the worst over a 25-day rollout.
- **Physics is a guardrail, not the engine**
  A light penalty (λ = 0.5 to 1) keeps rollout MAE below the plain MLP. Heavier physics (λ ≥ 10) drifts worse than no physics at all.

- **Sealed λ = 0.5**
  Inside the rollout-optimal band, chosen for the smaller worst-case swing.



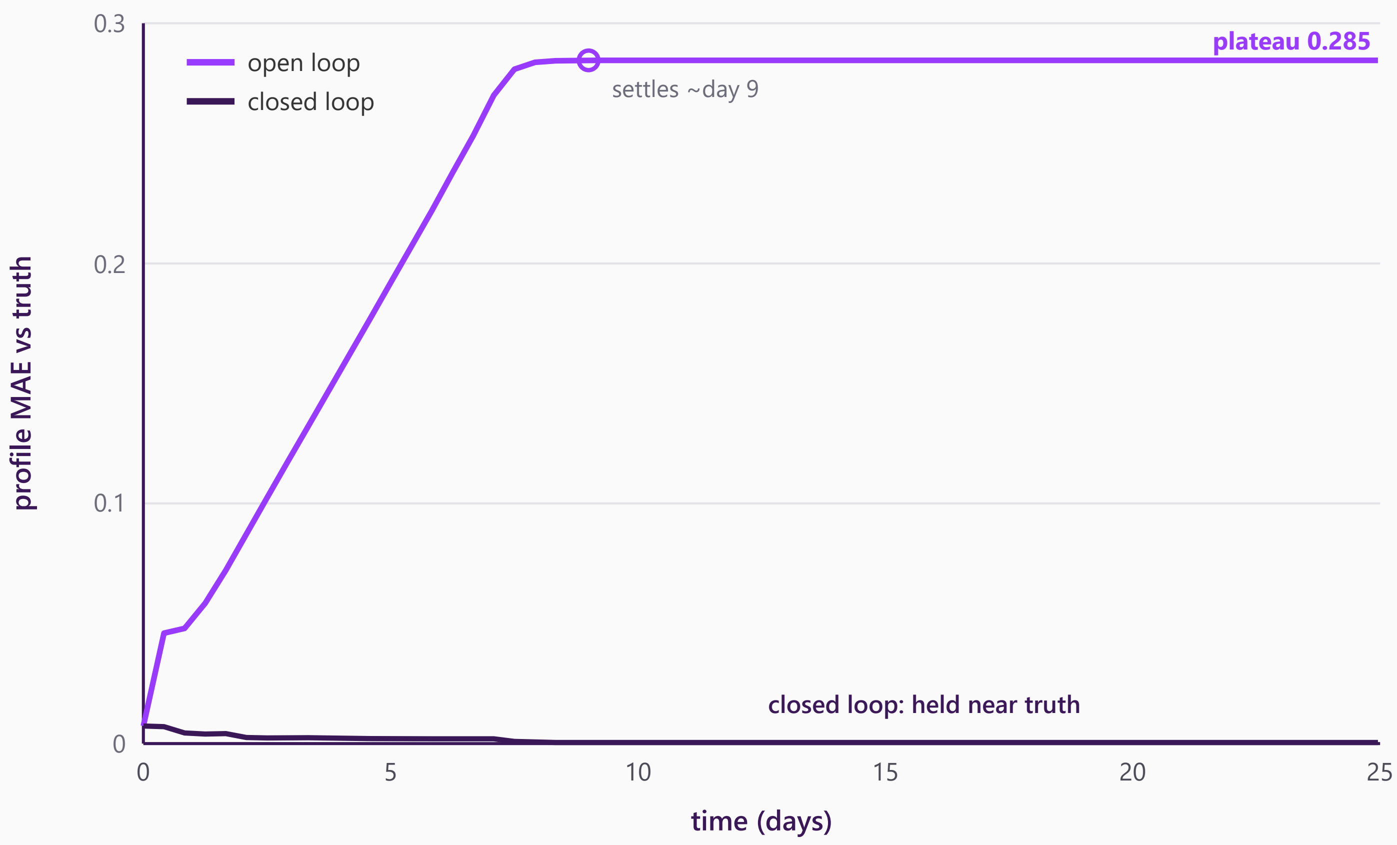
Y-axis: rollout MAE = mean $|Se_{pred} - Se_{true}|$ over the settled 25-day trajectory. B = 1, θ_i = 0.14, constant Q = 24. Source: S2 digest F36.

Open-loop result

Open-loop drift is a stress test, not the operating error

- **Open loop**
  Running on its own predictions for 25 days, the surrogate drifts to about 0.285 MAE by day 9, then plateaus.
- **Closed loop**
  Given the true saturation each step, the same model stays under 0.011 and settles near 0.004.

- **What we deploy**
  We run closed loop, so 0.004 is the operating error. The 0.285 open-loop number is only a stress test.



Open loop vs closed loop, true saturation fed back each step, constant Q = 24, θ_i = 0.14, one column, 25 days. Source: pgml_lam0p5.pt.

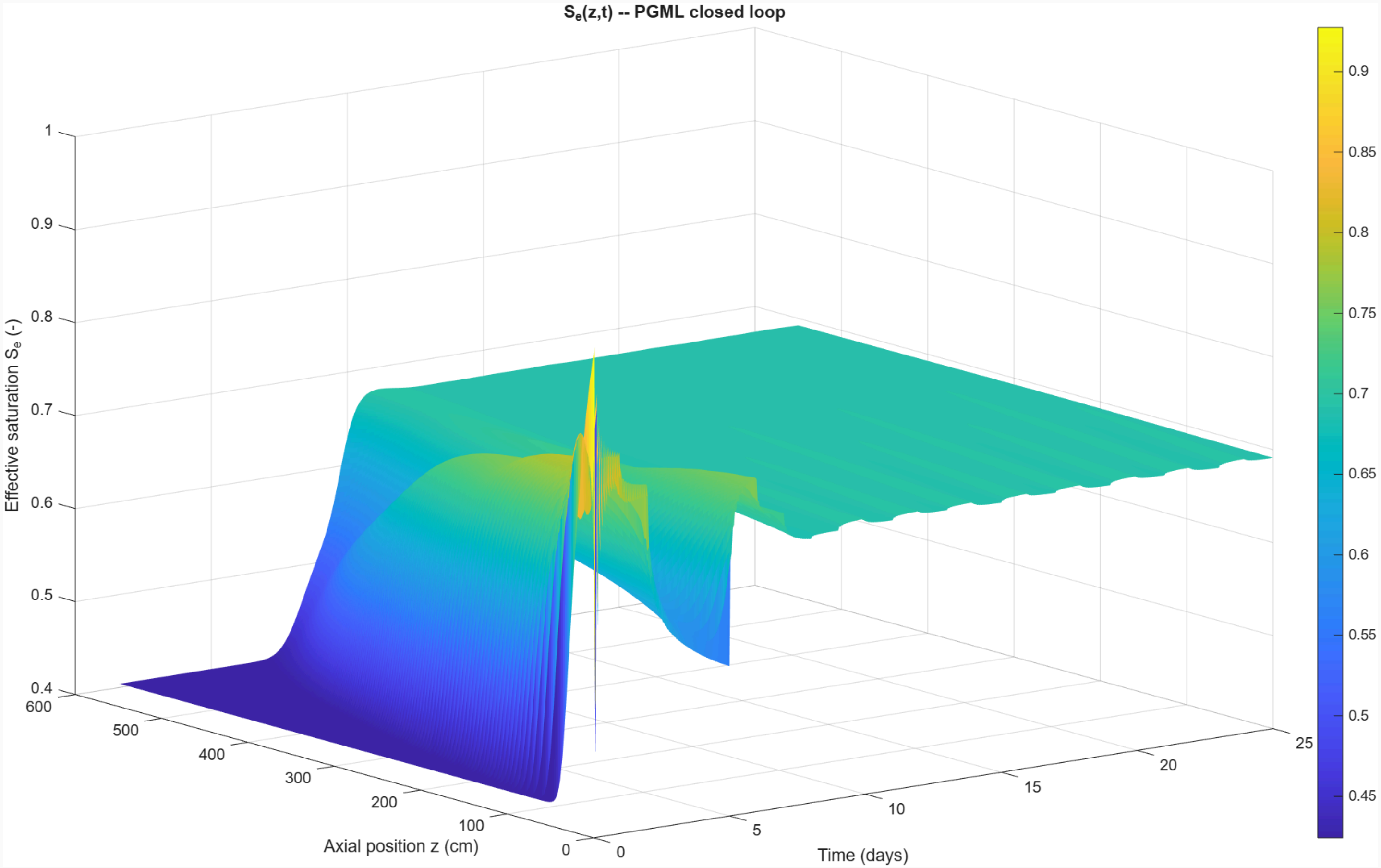
Closed-loop result

The surrogate drives a dry column to the setpoint and holds it

- **Surrogate as the controller model**
  Each step, sweep candidate daily R, roll the surrogate one day ahead, and take the R whose predicted profile sits closest to the setpoint.

- **Reaches and holds the setpoint**
  From a dry start near Se 0.42, it reaches 0.70 by about day 9 and holds at mean Se about 0.69.

- **Sealed result**
  A dry column driven to the setpoint and held for the rest of the run.



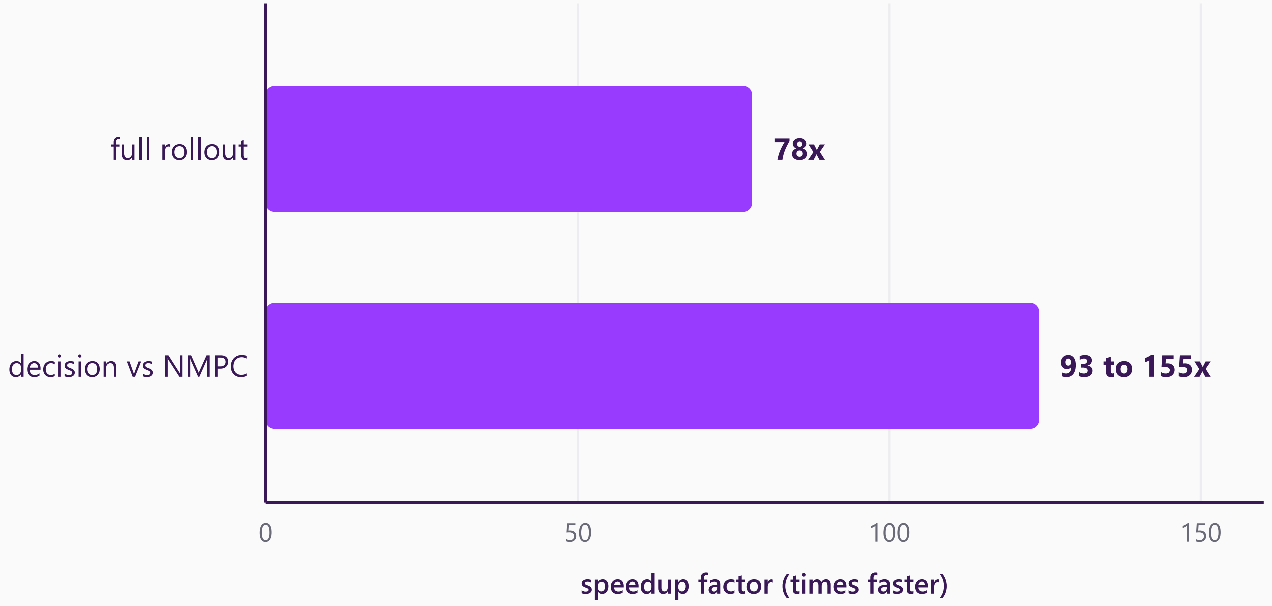
Se(z,t) over 25 days, PGML closed loop

Closed loop with the sealed PGML surrogate as the internal model, one column, setpoint 0.70. Source F43.

Speed

- **Full rollout**
  One 25-day trajectory from the surrogate against the simulator, same GPU.
- **Decision vs NMPC**
  One control decision in 0.65 ms.

- **The real win**
  Removing online identification, 1.4 to 3.5 s per step.



Rollout on the same GPU (RTX 5060). Decision vs NMPC is cross-hardware (RTX 5060 vs Ryzen 7 3750H CPU), so indicative. Source F35, F39.

Thank you

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